DAY 09 | HANDS-ON ACTIVITIES

FastAPI + MongoDB and JavaScript Basics: Problem Statements

Seven problems: four on MongoDB APIs, three in JavaScript. Run JavaScript with Node.js (in Colab: !node file.js).

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Seed and Paginate

Level: Warm-up | **Concepts:** insert_many, skip, limit

Seed 45 products. Write a function page(n, size) that returns one page plus total and total_pages. Print pages 1 and 5.

Sample run / expected result

```
page(5, 10) -> 5 items, total 45, total_pages 5
```

Hints

- total_pages = (total + size - 1) // size

Problem 2: Products API

Level: Easy | **Concepts:** FastAPI + PyMongo CRUD

Build POST, GET list (with page and size), GET one, PATCH and DELETE for products. Return 404 for invalid or unknown ids.

Sample run / expected result

```
GET /products/abc -> 404
GET /products?page=2&size=5 -> {"page": 2, "items": [...]}
```

Hints

- ObjectId.is_valid(pid)

Problem 3: Revenue Report

Level: Medium | **Concepts:** aggregation: $match, $group, $sort

Given orders with category, amount and status, return revenue and order count per category for paid orders only, highest first, as GET /stats/revenue.

Sample run / expected result

```
[{"category": "electronics", "revenue": 52000, "orders": 7}, ...]
```

Hints

- Use $project to rename _id to category.

Problem 4: Indexes

Level: Medium | **Concepts:** create_index, index_information

Create a compound index on category and price and a unique index on sku. Show the index list, and show that inserting a duplicate sku fails.

Sample run / expected result

```
Indexes: _id_, category_1_price_-1, sku_1
Duplicate sku -> DuplicateKeyError
```

Hints

- from pymongo.errors import DuplicateKeyError

Problem 5: JavaScript Warm-up

Level: Easy | **Concepts:** const, let, functions, template literals

Write functions to convert Celsius to Fahrenheit, check if a number is even, and build a greeting with a template literal. Print the results with console.log.

Sample run / expected result

```
37 C = 98.6 F
7 is odd
Hello, Aiko! Welcome to Day 9.
```

Hints

- const toF = (c) => c * 9 / 5 + 32;

Problem 6: Array Workout

Level: Medium | **Concepts:** map, filter, reduce, objects

Given an array of product objects, print the names of products under 500, the total stock value, and the most expensive product.

Sample run / expected result

```
Under 500: Pen, Mug
Stock value: 48300
Most expensive: Monitor
```

Hints

- reduce((max, p) => p.price > max.price ? p : max)

Problem 7: Group By in JS

Level: Stretch | **Concepts:** objects as maps, loops

Write groupBy(items, key) that returns an object of arrays, then use it to group orders by status and print the count per status.

Sample run / expected result

```
{ paid: 4, pending: 2, cancelled: 1 }
```

Hints

- Start with {} and push into acc[item[key]] ??= [].

Ganbatte (頑張って): do your best